

# PRATHMESH DESHKAR

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## SKILLS

**Languages & Frontend:** Java, TypeScript, JavaScript, HTML/CSS, React, Next.js, Vite, Tailwind CSS, Radix UI

**Backend & Database:** Java 21, REST API, JDBC, Gradle, JWT, MongoDB Atlas, GridFS, MySQL, PostgreSQL, Prisma

**DevOps & Tools:** Docker, Render, Vercel, Electron, GitHub Actions, VS Code, Git, Linux, Kali Linux

**Core CS:** Data Structures, Algorithms, OOP, Time & Space Complexity

## EXPERIENCE

### Software Development Research – Java (Independent Project)

Jan 2023 – Dec 2023

- Developed a **custom AES-192 encryption enhancement** in Java, improving cipher security characteristics.
- Conducted quantitative analysis using **entropy, avalanche effect, and execution time** to measure security vs. performance trade-offs.
- Applied OOP, software testing, and cryptographic evaluation; **published results in IJSRD**, demonstrating applied research and secure software design.

## PROJECTS

### Sentra – Encrypted File Sharing Platform [Github] [Live Link]

Jan 2026 – Apr 2026

- Built a **full-stack encrypted file-sharing platform** with React + TypeScript frontend and Java 21 backend, deployed on Render.
- Implemented **AES-256-GCM encryption with RSA key wrapping**, JWT auth via HttpOnly cookies, refresh tokens, and rate limiting.
- Used **MongoDB Atlas + GridFS** for encrypted file storage with inbox/outbox lifecycle, file expiry, and self-destruct.
- Dual-platform release: **Vite + Tailwind web app** and **Electron-based Windows desktop** with auto-update support.

**Technologies:** Java 21, React, TypeScript, Vite, Tailwind CSS, MongoDB Atlas, GridFS, AES-256-GCM, RSA, JWT, Electron, Docker, Render

### Recruvia – AI-Powered Resume Screening Platform [Github] [Live Link]

May 2026

- Built an **AI-powered resume screening engine** using **Gemini 2.0 Flash + Zod schema enforcement** to rank candidates against job descriptions with deep contextual understanding.
- Implemented **real-time scoring via Server-Sent Events (SSE)**; multi-format parsing (PDF, DOCX, TXT); 0–100 scores with skill match/miss badges and explanations.
- Persisted sessions with **PostgreSQL + Prisma ORM**; CSV export of ranked results; hardened against **prompt injection and MIME spoofing**.

**Technologies:** Next.js 15, React, TypeScript, Tailwind CSS v4, PostgreSQL, Prisma, Gemini 2.0 Flash, Zod, Framer Motion, Vercel

## EDUCATION

### Madhav Institute of Technology and Science

Gwalior, Madhya Pradesh

Bachelor of Technology – Information Technology

Expected: Jun 2027

CGPA: 8.04 / 10 (as of 6th Semester)

### Government Polytechnic College

Betul, Madhya Pradesh

Diploma in Computer Science

Completed: 2024

CGPA: 8.28 / 10

### Secondary School Certificate (10th)

Completed: 2021

CGPA: 8.98 / 10

## ACHIEVEMENTS

- Published peer-reviewed research paper: **“Assessing the Impact of Increased MixColumns on AES Encryption Security and Performance”** – IJSRD (International Journal of Scientific Research and Development).

## CERTIFICATIONS & TRAININGS

- Software Engineering Virtual Simulation – Walmart (2025):** Data structures, software architecture, relational databases, and data munging.
- Solutions Architecture Virtual Simulation – AWS (2025):** Designed scalable hosting on Elastic Beanstalk; documented functionality and cost breakdown.
- Ethical Hacking – IIT Jodhpur (2022):** Penetration testing, cryptography, and web security using Kali Linux, Metasploit, Nmap, and Burp Suite.